



UIC Student Enrollment Prediction Using Random Forest Models & ARIMA Projections

This project predicts student enrollment at UIC using machine learning and time-series models, providing insights for retention, planning, and long-term forecasting.

Background & Context

- Originated as a request from Office of Institutional Research (OIR)
- OIR had been using simple linear regression (5–10 year projections)
- Asked if more advanced methods (ARIMA, classification models) could improve accuracy
- Collaborations: UIC OIR, UIS colleagues, internal AITS data science team
- Now in third iteration: ARIMA + Exponential Smoothing + Random Forest

Project originated from OIR (Office of Institutional Research), led by Thomas Warfield

OIR had been using simple linear regression for 5–10 year projections

Requested more advanced methods → ARIMA for time-series, Random Forest for individual-level prediction

Collaboration with UIS (Brian Catherwood, ML.Net + C# model) → shared best practices, compared challenges

First version: ARIMA model, later rebuilt in KNIME

Second iteration: added Exponential Smoothing where ARIMA struggled

Parallel track: developed Random Forest models for student-level continuation predictions

Current state = third iteration, combining ARIMA + Exponential Smoothing + Random Forest for both long-term and individual-level forecasts

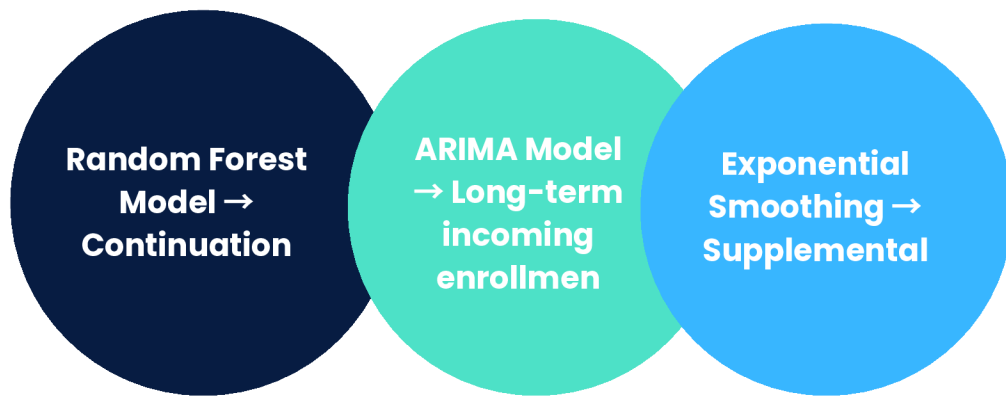
Project Goals

- Forecast long-term enrollment trends (ARIMA & Exponential Smoothing → 10-year projections)
- Aggregate results to college and department (scaling forecasts for planning)
- Predict at the individual level (Random Forest → student continuation)

Forecast long-term enrollment trends (10 years) using ARIMA and Exponential Smoothing → gives a strategic view of future student populations

Aggregate forecasts and student-level results to the college and department level → supports planning for faculty, courses, and budgets

Predict at the individual student level with the Random Forest model → identifies who is likely to return, supports retention, and builds the foundation for rollups



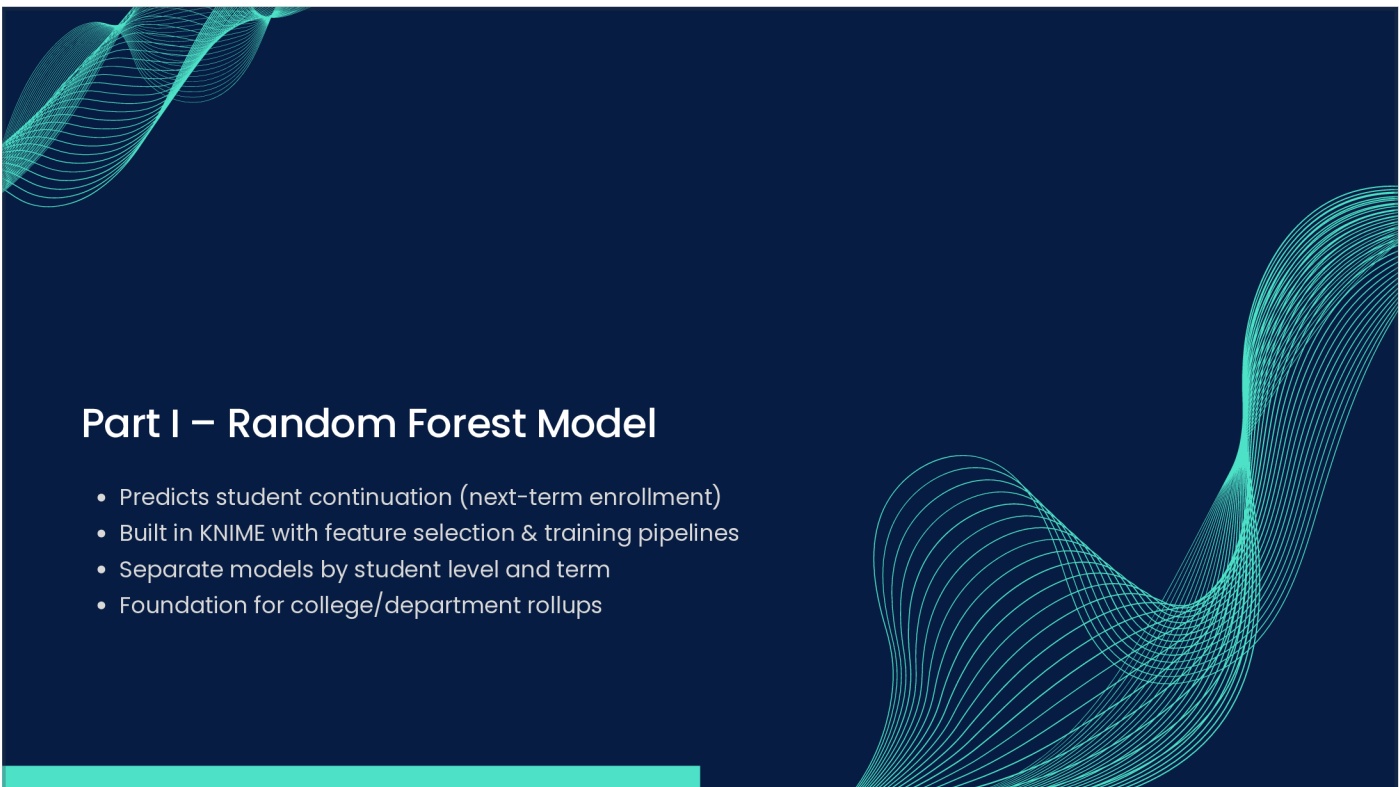
Models

Random Forest → individual continuation, core focus

ARIMA → 10-year forecasts, incoming students, colleges & levels

Exponential Smoothing → backup when ARIMA underperforms

Will dive deeper into each, main focus = Random Forest

The slide features a dark blue background with abstract teal wave-like patterns in the top-left and bottom-right corners. A solid teal horizontal bar is positioned at the bottom of the slide.

Part I – Random Forest Model

- Predicts student continuation (next-term enrollment)
- Built in KNIME with feature selection & training pipelines
- Separate models by student level and term
- Foundation for college/department rollups

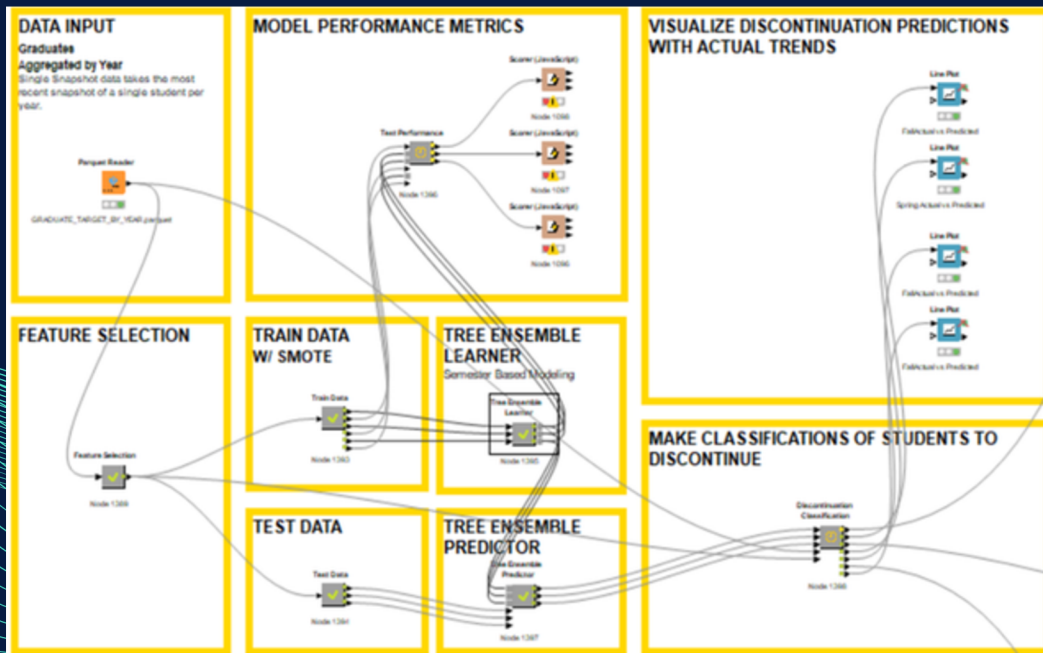
Predicts student continuation (next-term enrollment)

Built in KNIME → feature selection, balancing, training, validation

12 models total → by student level (UG, Grad, Prof, Law) and by term (Fall, Spring, Summer)

Individual-level predictions → identify at-risk students

Can be rolled up → college & department planning



Built in KNIME (open-source visual workflow tool for data science)

Data input via Parquet Reader (per student level)

Pre-selected feature columns from earlier Random Forest analysis

Training step: SMOTE node (balances discontinued vs continuing students) + Tree Ensemble Learner (built-in KNIME node)

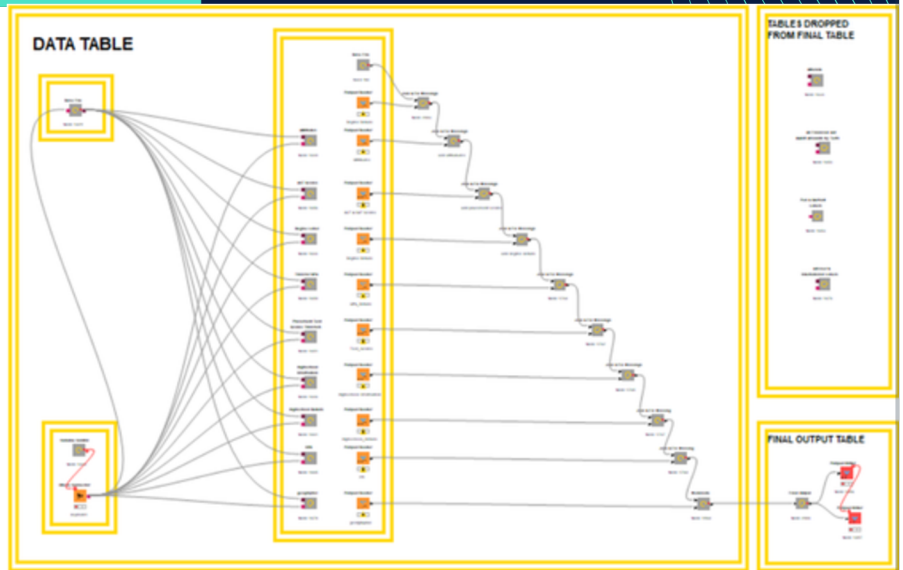
Performance metrics: accuracy, precision, recall (to check model fit during build)

Test data scored with Tree Ensemble Predictor (pre-built KNIME node)

Output: classifications (who discontinues) + actual vs predicted trend visualizations

Data Sources

- EDW: student registrations, admissions, demographics
- Additional details: degree info, attributes, GPA, test scores (ACT/SAT), high school, CRS, geography
- 15+ years of records
- Aggregated per student per year
- Initial pull included athlete, financial aid, Pell/Stafford, advisor, international labels → not approved for use



Broad EDW pull: registrations, admissions, demographics

Added: degree info, GPA, test scores, high school, CRS, geography

15+ years of records → aggregated per student/year

Feature selection → narrowed to correlated fields

Considered athlete, financial aid, Pell/Stafford, advisor, international → not approved

Started wide → narrowed to strong dataset

Target Variable

- Initial attempt: 3 outcomes → Continued, Graduated, Discontinued
- Final definition: Binary success vs. unsuccessful
- Successful = Continued + Graduated
- Unsuccessful = Discontinued

Attribute Selection | Tree Options | Ensemble Configuration | Flow Variables | Job Manager Selection | Memory Policy

Target Column: [S] SUCCESSFUL_OUTCOME_TARGET_BY_YEAR

Attribute Selection

☐ Use fingerprint attribute [no valid fingerprint input]

☒ Use column attributes

☒ Manual Selection ☐ Wildcard/Regex Selection

Exclude

Filter

No columns in this list

☒ Enforce exclusion

Include

Filter

- [S] DEPT_NAME
- [S] SEX_CD
- [S] PERS_MARITAL_STATUS_DESC
- [S] PERS_CITZN_TYPE_DESC
- [S] PERS_CITZN_TYPE_GROUP
- [S] RACE_ETH_DESC
- [S] IPEDS_RACE_ETH_DESC
- [D] AGE

☐ Enforce inclusion

Tried 3 outcomes: Continue, Graduate, Discontinue

Model struggled → poor accuracy/recall/precision

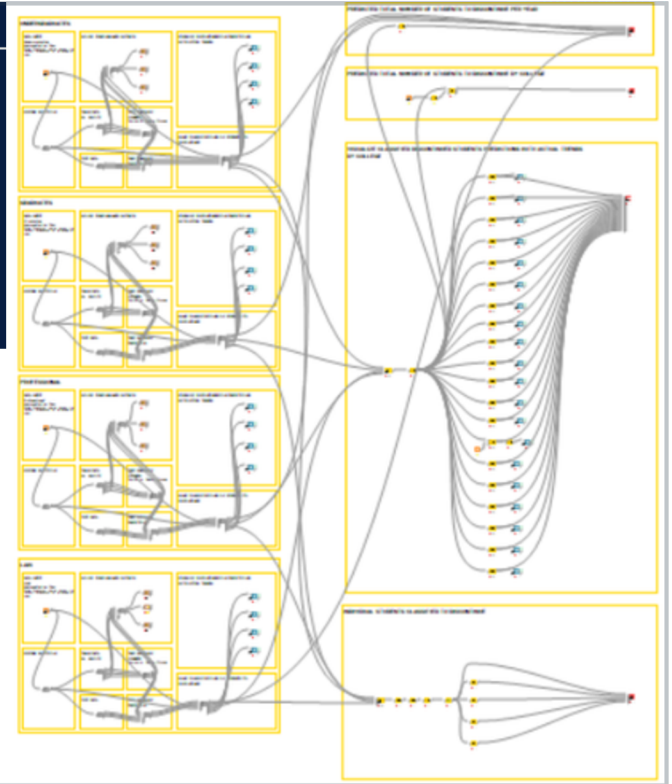
Switched to binary: Successful (Continue + Graduate) vs. Unsuccessful (Discontinue)

Simplification = major performance boost

Clear focus: predict student success vs. dropout

Segmentation

- By Student Level: Undergrad, Grad, Professional, Law
- By Term: Fall, Spring, Summer
- Seasonal patterns required separate models



Originally treated semesters the same → failed

Seasonal differences → major impact on continuation

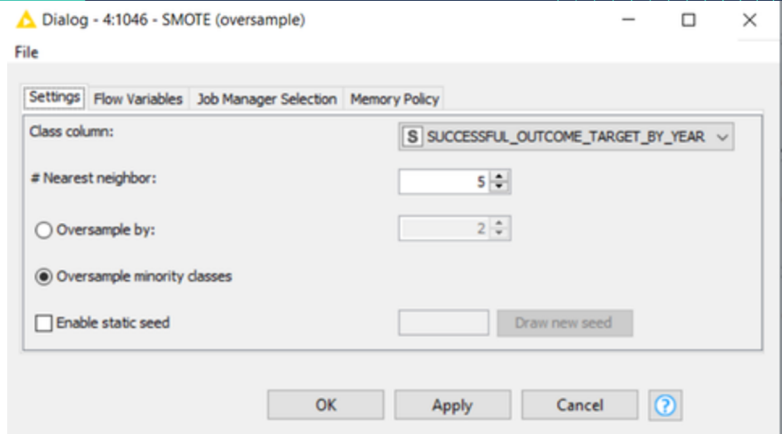
Fall = larger enrollments, Spring = distinct retention rates, Summer = consistently lower volume

Averaging didn't work → only solution = separate models

Segmented by level + term → 12 models total

Class Imbalance – SMOTE

- Discontinued students = minority class
- Model risk: bias toward “continued” predictions
- Solution: SMOTE oversampling → balance classes
- Improves ability to detect at-risk students



Imbalance: most students continue

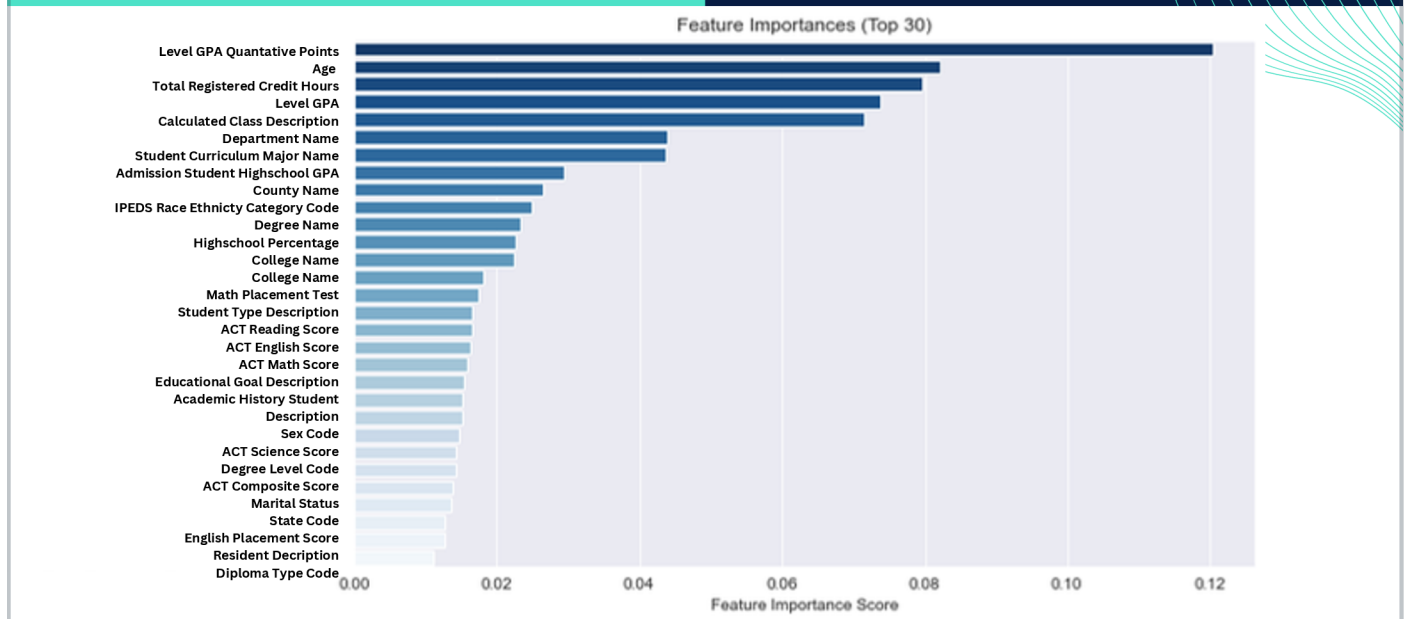
Risk: model predicts “continued” too often

Used SMOTE → synthetic discontinuations

Balanced data for training

Boosted detection of at-risk students

Feature Importance



Concise Cheat-Sheet Bullets (for you):

Python RF used for feature importance → static selector

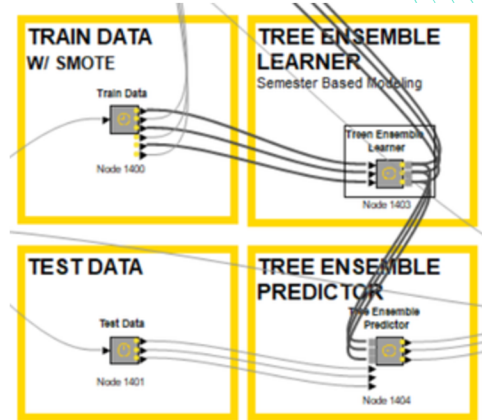
Locked features into KNIME model for consistency

Top predictors: GPA (level + HS), credit hours, age, class description, major

Clear academic/demographic signals of student success

Training Structure

- Balanced dataset using SMOTE
- Ensemble of 100 decision trees
- Split criterion: Information Gain Ratio
- Random sampling of rows (with replacement)
- Random subset of features per tree node



SMOTE → balanced dataset by creating synthetic “discontinued” cases so the model isn’t biased toward continuation

100 trees → enough trees to stabilize predictions without overloading computation

Info Gain Ratio → ensures splits give the most useful separation of students (better handling of categorical data)

Row sampling (with replacement) → each tree trains on slightly different subsets, increasing diversity and reducing bias

Random features per node → forces trees to consider different predictors, improving generalization and avoiding overfitting

Semester 3: Fall Terms

Scorer View

Confusion Matrix

Rows Number : 6991	0 (Predicted)	1 (Predicted)	
0 (Actual)	2787	19	99.32%
1 (Actual)	386	3799	90.78%
	87.83%	99.50%	

Class Statistics

Class	True Positives	False Positives	True Negatives	False Negatives	Recall	Precision	Sensitivity	Specificity	F-measure
0	2787	386	3799	19	99.32%	87.83%	99.32%	90.78%	93.23%
1	3799	19	2787	386	90.78%	99.50%	90.78%	99.32%	94.94%

Overall Statistics

Overall Accuracy	Overall Error	Cohen's kappa (κ)	Correctly Classified	Incorrectly Classified
94.21%	5.79%	0.882	6586	405

Accuracy: 94% → strong overall performance

Recall: 99% → catches almost all at-risk students

Precision: 88% → some over-flagging

Takeaway: Best term for identifying risk reliably

Semester 2: Spring Terms

Scorer View

Confusion Matrix

Rows Number : 44405	0 (Predicted)	1 (Predicted)	
0 (Actual)	1699	1775	48.91%
1 (Actual)	1590	39341	96.12%
	51.66%	95.68%	

Class Statistics

Class	True Positives	False Positives	True Negatives	False Negatives	Recall	Precision	Sensitivity	Specificity	F-measure
0	1699	1590	39341	1775	48.91%	51.66%	48.91%	96.12%	50.24%
1	39341	1775	1699	1590	96.12%	95.68%	96.12%	48.91%	95.90%

Overall Statistics

Overall Accuracy	Overall Error	Cohen's kappa (κ)	Correctly Classified	Incorrectly Classified
92.42%	7.58%	0.461	41040	3365

Accuracy: 92% → still solid overall

Recall: 49% → missed many at-risk students

Precision: 52% → low reliability

Takeaway: Weakest term, hardest to model

Semester 1: Summer Terms

Scorer View

Confusion Matrix

Rows Number : 10389	0 (Predicted)	1 (Predicted)	
0 (Actual)	872	197	81.57%
1 (Actual)	154	9166	98.35%
	84.99%	97.90%	

Class Statistics

Class	True Positives	False Positives	True Negatives	False Negatives	Recall	Precision	Sensitivity	Specificity	F-measure
0	872	154	9166	197	81.57%	84.99%	81.57%	98.35%	83.25%
1	9166	197	872	154	98.35%	97.90%	98.35%	81.57%	98.12%

Overall Statistics

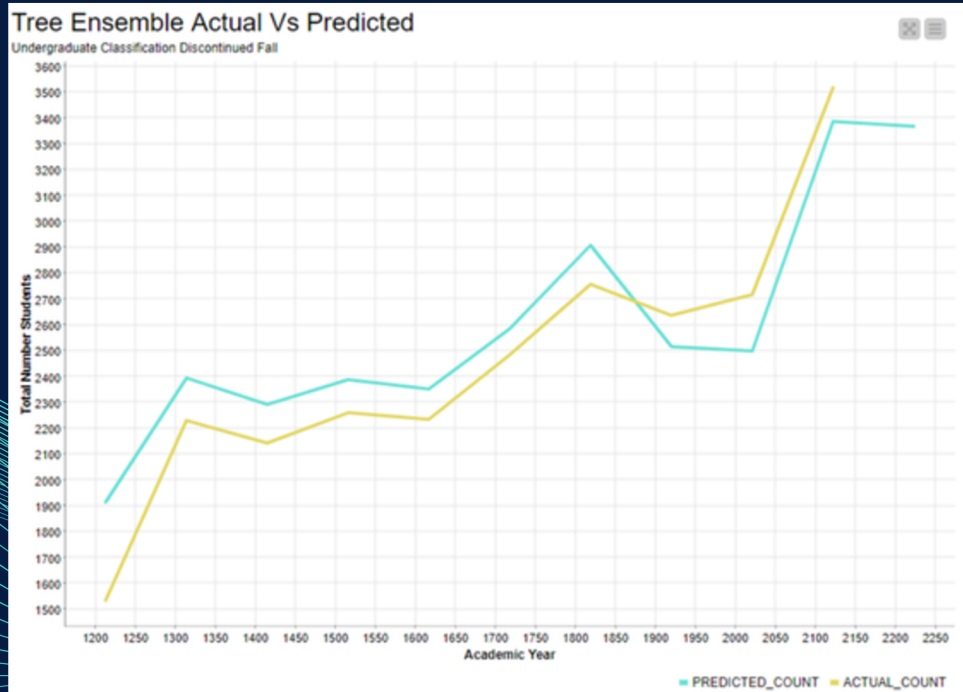
Overall Accuracy	Overall Error	Cohen's kappa (κ)	Correctly Classified	Incorrectly Classified
96.62%	3.38%	0.814	10038	351

Accuracy: 97% → highest overall

Recall: 82% → strong at-risk detection

Precision: 85% → predictions mostly correct

Takeaway: Most consistent and predictable term



Predicted vs. actual checked for each level + term

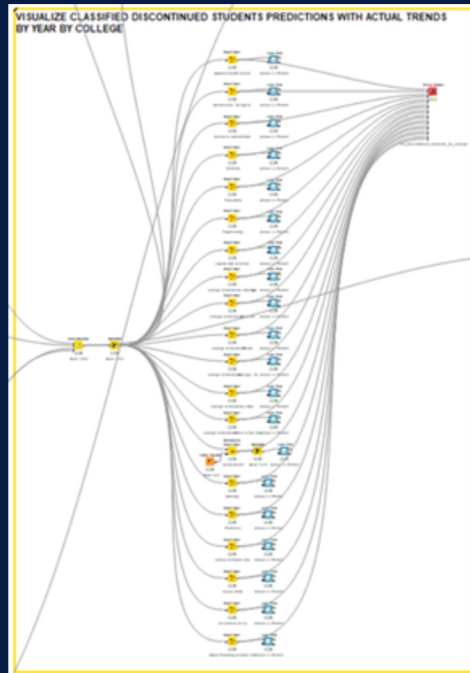
Example: Undergrad Fall = strong alignment

Captures rises/dips over time

Main takeaway: model tracks historical trends reliably

Transition to Validation

- Moving from model build → validation
- Predictions made in June
- Validation checked in October (Fall data)
- Compared predicted vs. actual continuation outcomes



June: made predictions

October: validated against actuals

Repolled data → compare predicted vs. actual

Main takeaway: test model on new, unseen data

Evaluation

- Predictions made June 2023 → validated in October 2023
- Compared predicted vs. actual student continuation
- Evaluation done in Python using standard metrics
 - For more information please refer to evaluation python script

Total students evaluated: 24,948
Correct predictions: 21,104
Accuracy: 84.6%
Precision: 87.9%
Recall: 94.1%
F1 Score: 90.9%

Main Takeaways:

- Model correctly classified ~85% of students.
- Very strong recall (94%) → captures most students at risk of discontinuation.
- Precision nearly 88% → most flagged students were truly at risk.
- Balanced performance (F1 = 90.9%) → reliable for both identifying and targeting retention interventions.

June: made predictions

October: validated against actuals

Repolled data → compare predicted vs. actual

Main takeaway: test model on new, unseen data

Evaluation – High Performance

	Program	Accuracy	Precision	Recall	F1 Score
0	Graduate Non-Degree Pharmacy	100%	100%	100%	100%
1	Graduate Non-Degree Public Health	100%	100%	100%	100%
2	Graduate Non-Degree Business Administration	100%	100%	100%	100%
3	Professional - Chicago Dentistry	97.0%	97.8%	99.2%	98.5%
4	Professional - Chicago College of Medicine Roc...	96.7%	96.7%	100%	98.3%
5	Professional - Chicago Pharmacy	96.0%	96.5%	99.5%	97.96%
6	Professional - Chicago College of Medicine (Ch...	95.8%	96.9%	98.8%	97.8%
7	Professional - Chicago Applied Health Sciences	94.9%	97.0%	97.7%	97.34%
8	Graduate - Chicago College of Medicine (Dean's...	98.6%	98.6%	100%	99.31%
9	Law – UIC School of Law	85.9%	92.9%	91.3%	92.1%

Flawless performance in some Graduate Non-Degree programs (Pharmacy, Public Health, Business Administration) with 100% across all metrics.

Professional programs (Dentistry, Medicine Rockford, Pharmacy, Applied Health Sciences) consistently show accuracy above 94%, with precision and recall both near or above 97%.

Graduate College of Medicine (Dean's Office) also demonstrates near-perfect reliability (Accuracy 98.6%, F1 99.3%).

Even the lowest of the top group – UIC School of Law – still performs strongly, with precision (92.9%) and recall (91.3%) supporting an overall F1 score of 92.1%.

Overall, these results highlight that well-structured, professional, and graduate-level programs are highly predictable with strong model stability.

Evaluation – Low Performance

	Program	Accuracy	Precision	Recall	F1 Score
0	Non-Credit – VP for Global Engagement	4.2%	4.2%	100%	8.0%
1	Undergrad Non-Degree LAS	72.8%	0%	0%	0%
2	Undergrad Non-Degree Engineering	54.0%	8.3%	66.7%	14.81%
3	Graduate Nursing (Non-Degree)	95.1%	100%	25%	40%
4	Undergrad Nursing	67.0%	73.4%	84.3%	78.47%
5	Graduate Business Administration	79.6%	82.9%	94.2%	88.18%
6	Graduate Pharmacy	81.9%	81.4%	100%	89.73%
7	Graduate LAS	87.3%	88.4%	98.4%	93.15%
8	Graduate Architecture/Arts	84.8%	84.8%	100%	91.8%
9	Graduate Social Work	86.6%	89.7%	96.1%	92.8%

Key Takeaways – Low Performing Models

Severe underperformance in Non-Credit (VP for Global Engagement):

Accuracy only 4.2%, Precision 4.2%, despite Recall at 100%.

Indicates extreme imbalance due to tiny sample size.

Undergrad Non-Degree LAS & Engineering:

LAS: 72.8% accuracy but 0% precision/recall, meaning the model fails to identify positives.

Engineering: 54% accuracy, Recall 66.7% but Precision only 8.3%, highlighting instability and poor predictive value.

Graduate Nursing (Non-Degree): High accuracy (95.1%) but very poor recall (25%), meaning the model misses most positive cases despite strong precision.

Undergrad Nursing: Accuracy 67%, precision and recall trade-off leads to modest predictive reliability (F1 ~78%).

Graduate-level programs (Business Admin, Pharmacy, LAS, Architecture/Arts, Social Work):

Accuracy ranges ~80–87%, with decent precision and recall (80–95%).

These are adequate but not top-tier, showing room for tuning and improvement

High Accuracy and Precision:

Coll of Med Office of the Dean:

Exceptional accuracy and precision (both 98.63%) highlight the model's robust performance in predicting student continuation for medical office-related programs.

Coll Medicine at Chicago - CS:

Demonstrates high accuracy (95.62%) and precision (96.71%), signifying reliable predictions for medicine-related programs at Chicago - CS.

Dentistry:

Strong accuracy (95.22%) and precision (95.88%) showcase the model's effectiveness in predicting student continuity for Dentistry programs

College-level results vary widely.

Some colleges achieved exceptionally high accuracy/precision (>95%).

Medical and health-related programs were most predictable.

Sets the stage to compare best vs. worst in upcoming slides.

Challenges and Opportunities for Improvement:

VP Undergrad Affrs & Acadmc Prgms:

Challenges are evident with 86.78% accuracy, where precision is notably absent. Given the small population size, it is reasonable to attribute the lower accuracy to this factor, and further investigation may not be necessary.

Undergrad Non-Degree Chicago:

Presents challenges with 76.41% accuracy, lower precision (6.59%), and recall (20.00%). The small population size is likely the main contributing factor to the lower accuracy, and this subset might not require extensive investigation beyond acknowledging this limitation.

Contextual Considerations:

Non-Credit – Chicago, VP for Global Engagement: Extreme cases with minimal samples and 4.17% accuracy emphasize

Some colleges/programs show weaker results.

VP Undergraduate Affairs: 86.8% accuracy, weaker precision.

Undergrad Non-Degree: 76% accuracy, precision 6.6%, recall 20% → hardest group.

Non-Credit/Global Engagement: very low accuracy (4%), extreme small samples.

Main takeaway: Model struggles with small population sizes, not core programs.

High Precision and Recall (Ideal)

Graduate Non-Degree Chicago_Pharmacy:

Accuracy:100%

Precision, Recall, F1 Score:100%

Assessment:The model excels in accurately predicting students likely to continue their studies in Pharmacy, with no false positives or false negatives. It's an ideal performance.

Graduate Non-Degree Chicago_School of Public Health:

Accuracy:100%

Precision, Recall, F1 Score:100%

Assessment: Similar to Pharmacy, the model performs flawlessly in predicting students who continue their studies in the School of Public Health.

Graduate Non-Degree Pharmacy → 100% accuracy, precision, recall, F1 (perfect predictions).

Graduate Non-Degree Public Health → 100% across all metrics as well.

Main takeaway: Model can perform flawlessly in small, stable, well-defined student groups.

Balanced Precision and Recall

Professional – Chicago_Dentistry:

Accuracy: 97.0%

Balanced Precision (97.8%) and Recall (99.2%)

Assessment: The model maintains a strong balance between precision and recall, suggesting a reliable identification of students continuing in Dentistry.

Professional – Chicago_College of Medicine Rockford:

Accuracy: 96.7%

Balanced Precision (96.7%) and Recall (100%)

Assessment: The model demonstrates good balance, effectively identifying students continuing in the College of Medicine Rockford.

Dentistry → 97% accuracy, ~98% precision, 99% recall → strong, balanced performance.

College of Medicine Rockford → 96.7% accuracy, 96.7% precision, 100% recall → captures all students at risk.

Main takeaway: Model is both precise and comprehensive in identifying continuation outcomes for professional programs.

Random Forest Wrap-Up

- Strong performance across most colleges/programs
- Flawless results in some well-defined, stable groups
- Challenges in small-population colleges (harder to predict)
- Identifies at-risk students at the individual level
- Flexible, auditable model → unlike proprietary black-box tools
- Room for refinement, strong potential for future use

Concise Talking Notes

Model performed well overall; strong, consistent results in many colleges.

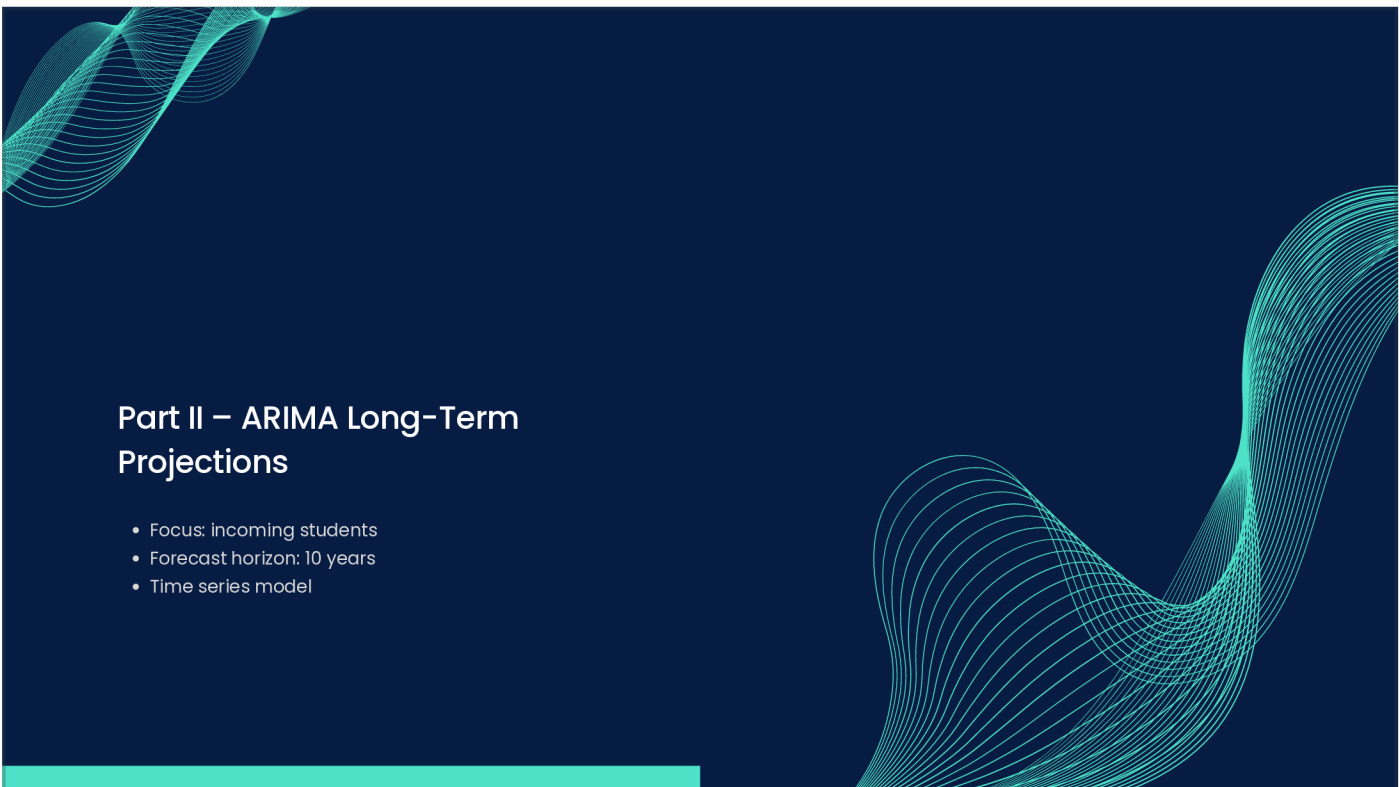
Flawless predictions in stable, well-defined programs.

Struggles appear in small-population colleges → harder to predict.

Key strength: identifies at-risk students down to the individual level.

Flexible and transparent — fully auditable, unlike proprietary black-box tools.

Solid foundation with room for refinement and future use.



Part II – ARIMA Long-Term Projections

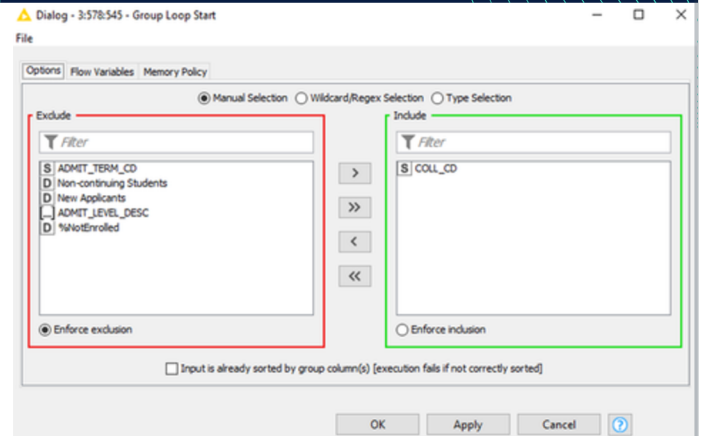
- Focus: incoming students
- Forecast horizon: 10 years
- Time series model

ARIMA Data

ARIMA (AutoRegressive Integrated Moving Average) is a time-series forecasting method. It's used to predict future values based on past data by combining three ideas:

- **AutoRegressive (AR):** Uses past values to predict current ones.
- **Integrated (I):** Applies differencing to make data more stable over time.
- **Moving Average (MA):** Uses past forecast errors to improve accuracy.

- 15+ years of applicant/enrollment history
- % Not Enrolled (admitted but didn't enroll)
- New Enrollments



Key Takeaways

15+ years of historical enrollment data used.

Focus on % not enrolled (yield gap) and new enrollments.

Data structured with clear include/exclude filtering for accuracy.

This is the dataset that powers our ARIMA forecasting.

We included long-term enrollment history to capture stable trends.

Key measures:

% of admitted students who don't enroll (yield gap).

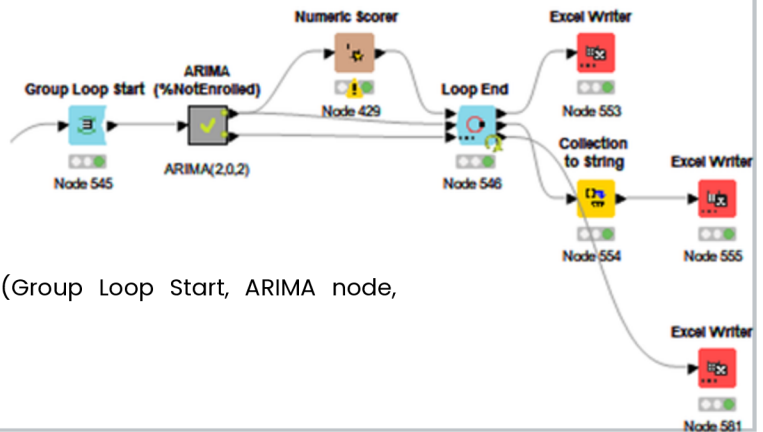
Number of new enrollments each cycle.

Data filtering ensures we're analyzing the right variables per college.

Sets the stage for accurate, institution-specific forecasting.

ARIMA Workflow

- ARIMA Workflow
- KNIME implementation
- Looped by college × student level
- Evaluated with R^2 , MAE, RMSE
- Workflow diagram showing KNIME nodes (Group Loop Start, ARIMA node, Numeric Scorer, Loop End, Excel Writers).



Built original ARIMA workflow in KNIME.

Looped predictions by college × student level.

Used R^2 , MAE, RMSE for evaluation:

R^2 (Coefficient of Determination) = measures overall model fit, showing how well the model explains the variance in the data.

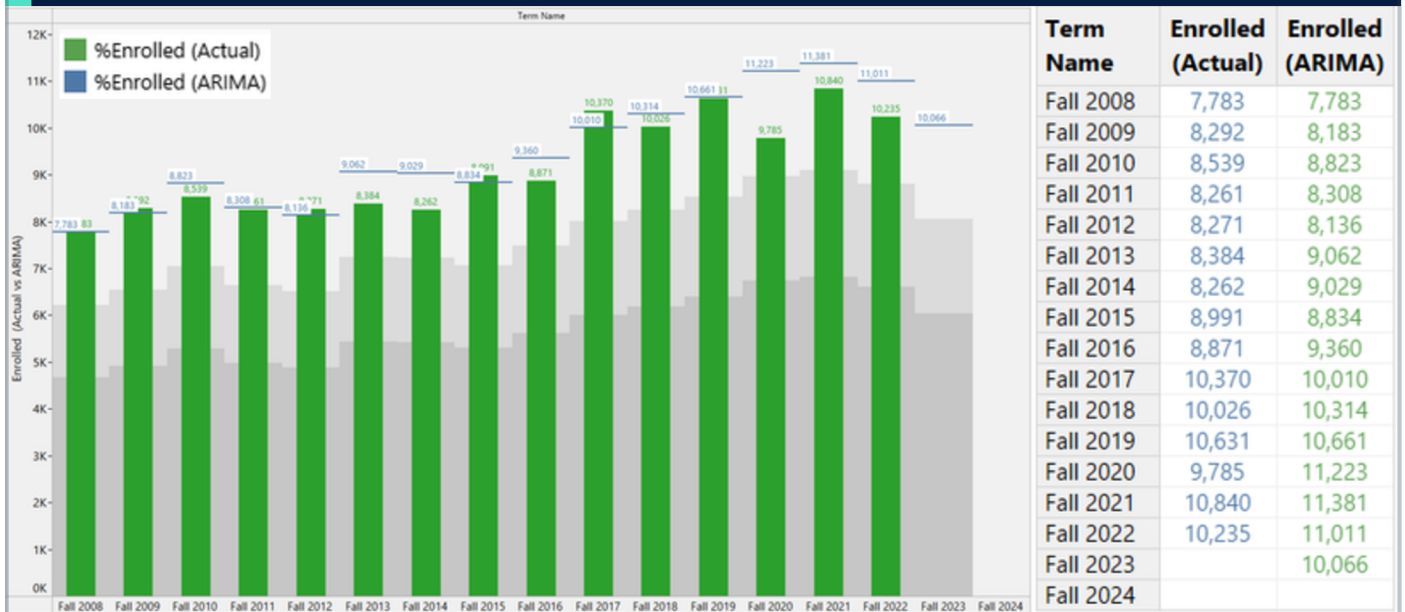
MAE (Mean Absolute Error) = represents the average size of prediction errors, regardless of direction.

RMSE (Root Mean Squared Error) = emphasizes larger errors by squaring them before averaging, making it more sensitive to big misses.

First KNIME version worked but had limitations.

Improved with exponential smoothing → later implemented in Python for flexibility.

ARIMA Predictions



This chart shows actual vs. ARIMA-predicted enrollment percentages across terms.

Covers more than a decade of data, with a 10-year projection window.

Green bars = actual enrollments, lighter overlay = ARIMA predictions.

The ARIMA model uses historical enrollment/admission patterns to estimate future enrollments.

Percent enrolled is used instead of raw counts because it's a normalized measure — it adjusts for fluctuations in admitted student pools and focuses on who actually enrolls.

Overall, the ARIMA model tracked trends well, with some underestimation in later years.

These results served as the baseline forecasting approach before introducing Exponential Smoothing refinements

ARIMA Predictions

- Using 15 years of Fall enrollment and admissions data at UIC, an ARIMA model was developed to predict the percentage of admitted students who would not enroll in the following two Fall terms.
- The performance metrics from this model are shown in the adjacent image.

- Strong overall model fit
- Very low prediction error
- Slight underestimation tendency
- Reliable for forecasting non-enrollment

R^2	0.74
mean absolute error	0.015
mean squared error	0
root mean squared error	0.02
mean signed difference	-0.011
mean absolute percentage error	0.024
adjusted R^2	0.697

Built ARIMA with 15 years of fall data.

$R^2 = 0.74 \rightarrow$ strong model fit.

MAE = 0.015, RMSE = 0.02 $\rightarrow$ very low prediction error.

Small bias: slight underestimation (-0.011).

MAPE = 2.4% $\rightarrow$ reliable forecasting accuracy.

Solid first iteration but limited.

Next: Exponential smoothing improved accuracy by selecting best-fit models at the college level $\rightarrow$ more granular 10-year projections, more actionable insights.



Part III – Exponential Smoothing Update

- ARIMA underperformed in some subsets
- Needed a supplemental method

Exponential Smoothing Approach

- ARIMA used for 10-year enrollment forecasting
- Some colleges showed poor fit (low R^2 , unstable trends)
- Exponential Smoothing added as fallback → improved stability
- Best-fit model selected per college for stronger projections
- Built originally in KNIME, later implemented in Python
- Data pipelines & transformations prepared internally

ARIMA baseline → 10-year forecasts

Added Exponential Smoothing where ARIMA struggled

Best-fit per college → stronger accuracy & stability

Ani = new enrollments; my role = continuing students

Used his KNIME workflow, adapted parameters/data

Later moved to Python for scalability



Questions